

LECTURE NOTES MATH255

SVEN SANDFELDT

1. LECTURE 22

In this class we prove that one can not, in general, solve a polynomial equation in radicals by applying the Galois theory developed in the previous three classes.

1.1. A necessary condition for not being able to solve a polynomial equation in radicals. Let $f(x) \in \mathbb{Q}[x]$. In this section we find a somewhat checkable condition to see if the zeros of $f(x)$ can be obtained from radicals of its coefficients (or more generally, from the rational numbers). Recall the following theorem from last class:

Theorem 1.1. *Let $\alpha \in \overline{\mathbb{Q}}$ be solvable by radicals, and let $f_\alpha(x) \in \mathbb{Q}[x]$ be the irreducible polynomial for α . If E_α is the splitting field for $f_\alpha(x)$ then $\text{Gal}(E_\alpha/\mathbb{Q})$ is a solvable group.*

To use the theorem, we will use an interpretation of Galois groups of splitting fields. Let $f(x) \in \mathbb{Q}[x]$ be a polynomial of degree n with zeros $\alpha_1, \dots, \alpha_n$ (we assume, for simplicity, that the zeros are distinct, this is always satisfied if $f(x)$ is irreducible). We can write the splitting field of $f(x)$ as $E_f = \mathbb{Q}(\alpha_1, \dots, \alpha_n)$. Any element $\psi \in \text{Gal}(E_f/\mathbb{Q})$ will map a zero of $f(x)$ to a zero of $f(x)$. That is, for every α_j there is some α_i such that $\psi(\alpha_j) = \alpha_i$. It follows that we can define a map $\sigma_\psi : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ by:

$$(1.1) \quad \psi(\alpha_j) = \alpha_{\sigma_\psi(j)}.$$

So we obtain a map $\Phi : \text{Gal}(E_f/\mathbb{Q}) \rightarrow S_n$ defined by $\Phi(\psi) = \sigma_\psi$. One checks that Φ is an injective homomorphism (why? Prove this! Injectivity follows since if $\psi \in \text{Gal}(E_f/\mathbb{Q})$ fix every α_j then ψ is the identity map). From this point of view we can always view $\text{Gal}(E_f/\mathbb{Q})$ as a subgroup of S_n . In the remainder of this section we will consider $f(x)$ of degree 5 and derive sufficient conditions to guarantee that $\text{Gal}(E_f/\mathbb{Q})$ is isomorphic to S_5 . Since S_5 is not a solvable group this will then imply that for any such polynomial the zeros are not solvable in radicals.

Lemma 1.1. *Let $f(x) \in \mathbb{Q}[x]$ be an irreducible degree 5 polynomial, then $\text{Gal}(E_f/\mathbb{Q})$ when viewed as a subgroup of S_5 contains a 5-cycle.*

Remark 1. This lemma does not hold for general n , it uses the fact that 5 happens to be prime (we will apply Cauchy's theorem in the proof).

Proof. In this proof, we will view $\text{Gal}(E_f/\mathbb{Q})$ as a subset of S_5 , so each $\psi \in \text{Gal}(E_f/\mathbb{Q})$ will be a map $\psi : \{1, \dots, 5\} \rightarrow \{1, \dots, 5\}$ (which is obtained by the relation $\psi(\alpha_j) = \alpha_{\sigma_\psi(j)}$, that is we identify ψ with σ_ψ). We consider the map $q : \text{Gal}(E_f/\mathbb{Q}) \rightarrow \{1, \dots, 5\}$ defined as $q(\psi) = \psi(1)$. Define a subgroup $H = \{\psi \in \text{Gal}(E_f/\mathbb{Q}) \mid \psi(1) = 1\}$ of $\text{Gal}(E_f/\mathbb{Q})$. Note that if $\psi \in H$ and $\eta \in \text{Gal}(E_f/\mathbb{Q})$ then $q(\eta\psi) = \eta\psi(1) = \eta(1) = q(\eta)$, so q descends to a map $\tilde{q} : \text{Gal}(E_f/\mathbb{Q})/H \rightarrow \{1, \dots, 5\}$. We claim that $\tilde{q}$ is injective. Indeed, if $\tilde{q}(\eta H) = \tilde{q}(\eta' H)$ then $\eta^{-1}\eta'(1) = 1$ which implies that $\eta^{-1}\eta' \in H$ or $\eta' \in \eta H$ which implies $\eta' H = \eta H$. We also claim that $\tilde{q}$ is surjective. To see this, note that for every $j \in \{1, \dots, 5\}$ there is some $\psi \in \text{Gal}(E_f/\mathbb{Q})$ such that $\psi(1) = j$ (why is this? What property does this translate to for the zeros of $f(x)$? This is where we use that $f(x)$ is irreducible, since this implies that for any α_j there is some $\psi \in \text{Gal}(E_f/\mathbb{Q})$ such that $\psi(\alpha_1) = \alpha_j$ since α_1 and α_j are conjugate). It follows that $\tilde{q}(\psi H) = j$. It follows that $\tilde{q}$ is a bijection so we have $|\text{Gal}(E_f/\mathbb{Q})/H| = 5$. By Lagrange's theorem we have $|\text{Gal}(E_f/\mathbb{Q})| = |H| \cdot |\text{Gal}(E_f/\mathbb{Q})/H| = 5|H|$ which implies that $|\text{Gal}(E_f/\mathbb{Q})|$ is divisible by 5. By Cauchy's theorem it follows that there is some $\psi \in \text{Gal}(E_f/\mathbb{Q})$ of order 5, this element corresponds to a 5-cycle in S_5 . ■

Lemma 1.2. *Let $f(x) \in \mathbb{Q}[x]$ be an irreducible polynomial of degree 5, and view $\text{Gal}(E_f/\mathbb{Q})$ as a subgroup of S_5 . If $\text{Gal}(E_f/\mathbb{Q})$ contains a transposition then:*

$$(1.2) \quad \text{Gal}(E_f/\mathbb{Q}) = S_5.$$

Proof. By the previous lemma we know that $\text{Gal}(E_f/\mathbb{Q})$ contains a 5-cycle. What we will prove here is a general fact of S_5 which is that if a subgroup $H \leq S_5$ contains a transposition and a 5-cycle then $H = S_5$.

Let $H \leq S_5$ be any subgroup of S_5 that contains a 5-cycle and a transposition. There is no loss of generality to assume that H contains the 5-cycle $(1, 2, 3, 4, 5)$, and some transposition $(1, n)$, $n \in \{2, 3, 4, 5\}$. Consider the transposition:

$$(1.3) \quad (1, \dots, 5)^{5-(6-n)}(1, n)(1, \dots, 5)^{6-n}$$

we can calculate this for $n = 2, 3, 4, 5$:

$$\begin{aligned} n = 2 : & (1, \dots, 5)^1(1, 2)(1, \dots, 5)^4 = (2, 3), \\ n = 3 : & (1, \dots, 5)^2(1, 3)(1, \dots, 5)^3 = (3, 5), \\ n = 4 : & (1, \dots, 5)^3(1, 4)(1, \dots, 5)^2 = (4, 2), \\ n = 5 : & (1, \dots, 5)^4(1, 5)(1, \dots, 5)^1 = (5, 4), \end{aligned}$$

and in each case we see that H must contain the element:

$$\begin{aligned} n = 2 : & (1, 2)(2, 3) = (1, 2, 3), \\ n = 3 : & (1, 3)(3, 5) = (1, 3, 5), \\ n = 4 : & (1, 4)(4, 2) = (1, 4, 2), \\ n = 5 : & (1, 5)(5, 4) = (1, 5, 4) \end{aligned}$$

which is a 3-cycle, so H always contains a 3-cycle. We claim that H also always contains a 4-cycle. For $n = 2$ this is immediate: $(1, 2, 3, 4, 5)(1, 2) = (1, 3, 4, 5)$. For $n = 5$ we note that $(1, 2, 3, 4, 5)(1, 5) = (2, 3, 4, 5)$, so in this case H also contains a

5-cycle. For $n = 3$ we note that $(3, 5)(1, 3)(3, 5) = (1, 5)$ so we obtain a 4-cycle as in the previous case. For $n = 4$ we have $(4, 2)(1, 4)(4, 2) = (1, 2)$ so we obtain a 4-cycle as in the case $n = 2$. In either case we have that H must contain 5-cycle, and 4-cycle, and a 3-cycle. By Lagrange's theorem it follows that $5 \cdot 4 \cdot 3 = 60$ divides $|H|$. We then have two possibilities, either $H = A_5$ in which case $|H| = 60$, but this is impossible since H contains a transposition by assumption and the sign of a transposition is odd. The only remaining possibility is that $H = S_5$. ■

1.2. A polynomial with non-solvable Galois group. In this section we produce a polynomial $f(x)$ of degree 5 whose Galois group is S_5 , and therefore not solvable. We will consider the polynomial:

$$(1.4) \quad f(x) = x^5 - 4x + 2.$$

Note that -4 and 2 are divisible by the prime 2 , and 2 is not divisible by $2^2 = 4$, so by Eisenstein's criterion we know that $f(x)$ is irreducible over $\mathbb{Q}$. Moreover, note that:

$$(1.5) \quad \lim_{x \rightarrow -\infty} f(x) = -\infty,$$

$$(1.6) \quad f(-1) = -1 + 4 + 2 = 5 > 0,$$

$$(1.7) \quad f(1) = 1 - 4 + 2 = -1 < 0,$$

$$(1.8) \quad \lim_{x \rightarrow \infty} f(x) = \infty,$$

so by the intermediate value theorem $f(x)$ must have a zero in $(-\infty, -1)$, $(-1, 1)$, and $(1, \infty)$. That is, $f(x)$ has at least 3 real zeros. We claim that $f(x)$ has precisely 3 real zeros. One way to see this is to note that $f(x)$ must either have 3 real zeros or 5 real zeros (since $f(x)$ has real coefficients so the non-real zeros come in complex conjugate pairs), and if $f(x)$ has 5 real zeros then the derivative $f'(x)$ must have 4 real zeros (why is this? Try to picture what the graph of $f(x)$ must look like if it has 5 zeros). However, we can calculate the zeros of $f'(x)$ as:

$$(1.9) \quad 0 = f'(x) = 5x^4 - 4, \quad x \in \left\{ \left(\frac{4}{5}\right)^{1/4}, -\left(\frac{4}{5}\right)^{1/4}, i\left(\frac{4}{5}\right)^{1/4}, i\left(\frac{4}{5}\right)^{1/4} \right\},$$

and since only two of these zeros are real we conclude that $f(x)$ must have precisely 3 real zeros. Notice that this calculation shows that $f(x)$ has 5 zeros which we write as $\{\lambda_1, \lambda_2, \lambda_3, s + i\tau, s - i\tau\}$ with $\lambda_1, \lambda_2, \lambda_3, s, \tau \in \mathbb{R}$ (that is, $f(x)$ has 3 real zeros and two non-real zeros that are complex conjugates).

Theorem 1.2. *Let E_f be the splitting field of $f(x)$, then $\text{Gal}(E_f/\mathbb{Q}) \cong S_5$. In particular, the zeros of $f(x)$ are not solvable by radicals, so there can be no formula in terms of radicals to obtain the zeros of $f(x)$ from the coefficients of $f(x)$.*

Proof. We begin by noting that the last claim of the theorem follows since the coefficients of $f(x)$ are rational numbers. The fact that the zeros of $f(x)$ can not be solvable by radicals is that S_5 is not solvable (why? Recall that A_5 is simple), Theorem 1.1, and the fact that $f(x)$ is the irreducible polynomial for any of its zeros.

By Lemma 1.2 it suffices to prove that when we view $\text{Gal}(E_f/\mathbb{Q})$ as a subgroup of S_5 then $\text{Gal}(E_f/\mathbb{Q})$ contains a transposition. We define the map $\psi : E_f \rightarrow E_f$ by $\psi(\alpha) = \bar{\alpha}$ (where $\bar{\alpha}$ is the complex conjugate of α), note that if $\psi(q) = q$ for any $q \in \mathbb{Q}$ and if α is a zero of $f(x)$ then so is $\bar{\alpha}$ since $f(x)$ has real coefficients, so this is indeed a map from E_f to E_f . Moreover, ψ fix every element of $\mathbb{Q}$ and from properties of complex conjugates we have $\psi(\alpha + \beta) = \psi(\alpha) + \psi(\beta)$ and $\psi(\alpha\beta) = \psi(\alpha)\psi(\beta)$. That is, $\psi \in \text{Gal}(E_f/\mathbb{Q})$. Moreover, we have $\psi(\lambda_j) = \lambda_j$ for $j = 1, 2, 3$ and $\psi(s + i\tau) = s - i\tau$. So ψ fix the 3 of the zeros, and interchanges the two non-real zeros. But that means that ψ corresponds to a transposition and the theorem follows. ■

Email address: sandfeldt@uchicago.edu